

ST LAWRENCE, NL, Canada

WMO: 711100

Lat: 46.920N

Lon: 55.380W

Elev: 159

StdP: 14.61

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.4	12.5	3.5	6.6	10.5	6.5	7.7	13.5	50.4	24.2	47.3	23.2	12.9	310	0.795

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	73.6	66.3	70.8	64.8	68.4	63.6	68.5	71.3	66.4	68.9	64.7	67.0	12.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.4	101.5	69.7	65.5	94.6	67.6	63.8	89.2	65.9	32.9	71.5	31.2	68.8	29.9	67.1	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.5	35.2	30.1	3.9	78.9	3.0	3.6	1.8	81.5	0.0	83.5	-1.7	85.6	-3.9	88.2		
(5)			3.6	70.1	3.0	2.9	1.4	72.2	-0.4	73.9	-2.1	75.6	-4.3	77.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	26.9	25.7	29.4	36.3	44.1	51.3	59.6	62.0	56.2	47.5	39.8	31.8	(6)	
	DBStd	13.58	6.77	7.22	6.37	4.87	4.57	4.85	4.46	4.22	5.42	5.87	6.14	7.02	(7)	
	HDD50	3681	717	680	639	410	191	39	0	0	11	121	309	564	(8)	
	HDD65	8183	1182	1100	1104	860	647	411	175	109	268	543	755	1029	(9)	
	CDD50	1001	0	0	0	0	9	79	297	373	197	43	3	0	(10)	
	CDD65	28	0	0	0	0	0	0	7	17	4	0	0	0	(11)	
	CDH74	78	0	0	0	0	1	5	25	31	16	0	0	0	(12)	
	CDH80	7	0	0	0	0	0	0	2	2	3	0	0	0	(13)	
(14) Wind	WSAvg	13.8	17.2	17.8	16.1	14.6	12.4	11.1	9.6	10.4	11.9	13.7	14.9	16.6	(14)	
(15) Precipitation	PrecAvg	63.1	5.6	5.2	5.1	4.9	4.6	5.6	4.3	5.1	5.8	6.1	5.8	5.4	(15)	
	PrecMax	78.2	8.5	9.7	8.2	9.4	8.0	13.7	7.8	16.1	10.0	10.7	10.2	10.7	(16)	
	PrecMin	48.0	2.3	1.7	2.9	2.2	2.3	2.0	1.8	2.1	2.2	1.6	3.2	3.0	(17)	
	PrecStd	7.5	1.4	2.2	1.3	2.2	1.3	2.6	1.6	3.0	1.8	2.7	1.6	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.2	44.9	47.4	53.8	66.5	72.5	77.0	77.3	74.7	65.5	56.8	51.2	(19)	
		MCWB	45.7	44.0	45.0	46.4	55.4	61.6	67.4	68.1	66.6	62.2	55.9	50.9	(20)	
	2%	DB	41.6	41.2	42.8	48.9	59.5	66.8	73.0	73.9	69.0	61.9	53.8	47.8	(21)	
		MCWB	41.2	40.6	40.5	44.4	50.9	58.4	65.2	67.3	64.1	59.2	53.0	47.1	(22)	
	5%	DB	38.8	38.5	39.9	46.1	55.6	63.2	70.0	71.3	66.3	59.1	51.9	44.9	(23)	
		MCWB	38.3	37.8	38.2	42.8	48.7	56.3	63.7	66.0	62.8	57.0	51.2	44.2	(24)	
	10%	DB	36.3	35.7	37.8	43.7	52.4	59.9	67.2	69.0	64.4	56.8	49.4	42.0	(25)	
		MCWB	35.6	34.6	36.3	41.1	47.5	54.6	62.5	64.8	61.5	54.7	48.4	41.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.0	44.4	45.5	48.5	57.7	64.2	71.4	73.0	69.1	64.0	55.5	51.1	(27)	
		MCDB	46.1	44.6	46.9	50.9	63.4	68.6	74.3	74.8	72.8	64.3	55.7	51.2	(28)	
	2%	WB	41.2	40.6	41.2	45.6	53.1	60.3	67.4	69.3	65.9	60.1	53.1	47.4	(29)	
		MCDB	41.4	41.0	42.0	47.5	56.9	64.8	70.5	71.6	68.1	61.3	53.4	47.7	(30)	
	5%	WB	38.4	37.8	38.7	43.5	50.4	57.8	65.1	67.3	64.0	57.5	51.1	44.2	(31)	
		MCDB	38.7	38.2	39.6	45.4	53.6	61.3	67.9	69.7	65.6	58.8	51.6	44.8	(32)	
	10%	WB	35.7	34.8	36.4	41.5	48.1	55.6	63.4	65.6	62.2	54.8	48.6	41.5	(33)	
		MCDB	36.1	35.5	37.5	43.3	51.5	58.8	65.9	67.8	63.8	56.3	49.2	42.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	11.0	11.1	12.0	14.2	13.9	12.2	12.0	13.3	12.1	10.8	10.3	(35)	
		MCDBR	13.7	13.3	12.7	15.6	20.5	20.2	17.5	15.6	14.4	13.2	12.7	13.0	(36)	
		MCWBR	14.3	13.4	11.1	11.8	13.1	12.9	11.4	10.3	10.6	11.1	12.6	13.3	(37)	
	5% WB	MCDBR	13.8	13.2	12.5	14.1	16.8	16.5	13.4	12.2	12.4	12.2	12.2	13.0	(38)	
		MCWBR	14.4	13.7	11.9	11.7	12.2	12.0	10.1	9.4	10.5	11.5	12.6	13.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.322	0.321	0.335	0.369	0.377	0.400	0.430	0.403	0.370	0.354	0.327	0.314	(40)	
	taud		2.349	2.426	2.428	2.358	2.395	2.355	2.318	2.398	2.468	2.456	2.460	2.383	(41)	
	Ebn at Noon		239	270	283	281	282	275	264	267	267	253	238	228	(42)	
	Edh at Noon		23	27	31	36	36	38	39	35	30	26	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		311	545	927	1334	1599	1631	1549	1382	1138	708	392	262	(44)	
	RadStd		25	50	54	114	123	168	109	88	71	51	50	22	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page